40 MHz clock rate, this provides a maximum IIR throughput of 1.8M to 3.1M samples/second. These IIR values are also shown in Fig. 28-11.

Next we come to the frequency domain techniques, based on the Fast Fourier Transform. FFT subroutines are almost always provided by the manufacturer of the DSP. These are highly-optimized routines written in assembly. The specification sheet of the ADSP-21062 SHARC DSP indicates that a 1024 sample complex FFT requires 18,221 clock cycles, or about 0.46 milliseconds at 40 MHz. To calculate the throughput, it is easier to view this as 17.8 clock cycles per sample. This "per-sample" value only changes slightly with longer or shorter FFTs. For instance, a 256 sample FFT requires about 14.2 clock cycles per sample, and a 4096 sample FFT requires 21.4 clock cycles per sample. Real FFTs can be calculated about 40% faster than these complex FFT values. This makes the overall range of all FFT routines about 10 to 22 clock cycles per sample, corresponding to a throughput of about 1.8M to 3.3M samples/second.

FFT convolution is a fast way to carry out FIR filters. In a typical case, a 512 sample segment is taken from the input, padded with an additional 512 zeros, and converted into its frequency spectrum by using a 1024 point FFT. After multiplying this spectrum by the desired frequency response, a 1024 point Inverse FFT is used to move back into the time domain. The resulting 1024 points are combined with the adjacent processed segments using the overlap-add method. This produces 512 points of the output signal.

How many clock cycles does this take? Each 512 sample segment requires two 1024 point FFTs, plus a small amount of overhead. In round terms, this is about a factor of five greater than for a single FFT of 512 points. Since the real FFT requires about 12 clock cycles per sample, FFT convolution can be carried out in about 60 clock cycles per sample. For a 2106x SHARC DSP at 40 MHz, this corresponds to a data throughput of approximately 660k samples/second.

Notice that this is about the same as a 60 coefficient FIR filter carried out by conventional convolution. In other words, if an FIR filter has less than 60 coefficients, it can be carried out faster by standard convolution. If it has greater than 60 coefficients, FFT convolution is quicker. A key advantage of FFT convolution is that the execution time only increases as the logarithm of the number of coefficients. For instance a 4,096 point filter kernel only requires about 30% longer to execute as one with only 512 points.

FFT convolution can also be applied in two-dimensions, such as for image processing. For instance, suppose we want to process an 800×600 pixel image in the frequency domain. First, pad the image with zeros to make it 1024×1024. The two-dimensional frequency spectrum is then calculated by taking the FFT of each of the rows, followed by taking the FFT of each of the resulting columns. After multiplying this 1024×1024 spectrum by the desired frequency response, the two-dimensional Inverse FFT is taken. This is carried out by taking the Inverse FFT of each of the rows, and then each of the resulting columns. Adding the number of clock cycles and dividing by the